



SAPIENZA
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Title

QUANTUM COMPUTING NOTES

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1 An Introduction to Quantum Computing

1.1 Debunking Quantum Computing Myths

- **Myth 1: Quantum computers will solve NP-complete problems efficiently.**
 - *Fact:* **False**, No efficient classical or quantum algorithm exists for NP-complete problems, and radical theoretical breakthroughs would be required to change this.
- **Myth 2: Quantum computers are too complicated to ever be built.**
 - *Fact:* **False**, Working Noisy Intermediate-Scale Quantum (NISQ) computers are operational today (e.g., systems built by IBM, Quantinuum, QuEra), including IBM's 1,121-qubit processor.
- **Myth 3: Quantum computing is just another fast hardware architecture (like GPUs or multi-core CPUs).**
 - *Fact:* **False**, It exploits quantum phenomena with no classical equivalent that cannot be simulated efficiently on classical machines, operating under a fundamentally different model of computation.
- **Myth 4: Quantum computing is hard (and requires knowing quantum physics).**
 - *Fact:* **Partially true**, but the mathematical foundation (formalized by John von Neumann) relies on complex linear algebra; the main current challenge is building proper high-level programming abstractions.
- **Myth 5: Quantum computers and classical computers solve the same problems.**
 - *Fact:* **True and False**, in terms of computability (they can simulate each other); however, classical simulation of quantum physics is exponentially slow, whereas quantum algorithms offer exponential speedups **for specific tasks**.

1.2 Brief History of Quantum Computing

- **1900:** Max Planck studies black-body radiation and discovers that electromagnetic radiation energy is emitted in discrete “quanta”.
- **1905:** Albert Einstein explains the photoelectric effect by proposing that light propagates as massless particles called photons.

- **1926:** Erwin Schrödinger formulates the time-dependent ($i\hbar \frac{d}{dt}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$) and time-independent ($\hat{H}|\Psi\rangle = E|\Psi\rangle$) equations describing the evolution of isolated quantum systems.
- **1927:** Werner Heisenberg introduces the uncertainty principle ($\sigma_x\sigma_p \geq \frac{\hbar}{2}$).
- **1932:** John von Neumann publishes *Mathematical Foundations of Quantum Mechanics*, establishing the theoretical foundation for quantum computing.
- **1963 / 1973:** Yves Lecerf and Charles Bennett introduce reversible Turing machines, proving that isolated quantum systems must undergo reversible computation.
- **1980 / 1982:** Paul Benioff, Yuri Manin, and Richard Feynman propose the first concepts of quantum computers to simulate physical quantum systems.
- **1994:** Peter Shor formulates the quantum factoring algorithm providing an exponential speedup over classical algorithms.
- **1996:** Lov Grover develops the quantum search algorithm providing a quadratic speedup over classical search.

1.3 Current State of the Industry

1.3.1 Sector Investments

- **Public Sector:** Substantial investments in quantum technologies, including the UK (~£1bn, highlighted in AUKUS), the EU (€1bn Quantum Flagship), the US, Germany/Bavaria (hundreds of millions of euros), and China (field leader in scientific journal publications).
- **Private Sector:** Commercial hardware developers selling systems (IBM, IonQ, IQM, Pasqal, QuEra, Quantinuum), private hardware initiatives (Google, Intel), research teams (Microsoft), cloud integrations (Amazon at Caltech), and financial applications in Quantum Machine Learning and simulation.

1.3.2 Technological Achievements

- **October 2019 (Google):** Demonstrated quantum supremacy using the 54-qubit *Sycamore* processor.
- **2023 (IBM):** Made 156-qubit systems publicly available and introduced the 1,121-qubit chip.
- **December 2023 (Harvard/MIT/QuEra):** Demonstrated error-corrected quantum algorithms on 48 logical qubits created from ~300 physical qubits

- **April 2024 (Microsoft & Quantinuum):** Created 4 highly reliable logical qubits from 30 physical qubits, achieving $\sim 99.8\%$ fidelity and an 800x reduction in error rate.
- **December 2024 (Google):** Achieved quantum error correction below the surface code threshold on the 105-qubit *Willow* system (executed random circuit sampling benchmark in under 5 minutes vs 10^{25} years required by supercomputers).
- **February 2025 (Microsoft):** Announced the *Majorana 1* topological chip utilizing Majorana particles for fault-tolerant quantum computing.
- **September 2025 (Harvard):** Neutral atom progress enabling the creation of 30,000 initialized qubits per second and coherent storage of 3,000 qubits for 2 hours.

2 Development

3 Conclusions

References

- [1] Leslie Lamport. *LaTeX: a Document Preparation System*. Addison Wesley, Massachusetts, 2 edition, 1994.